

Part 2 of 2

TUTORIAL ON ATMEGA32 DEVELOPMENT BOARD



■ SAGAR G. YADAV

Part 1 of this article in March issue discussed five applications of the AVR ATmega32 microcontroller. Here we explore some more features of this microcontroller, including the external interrupt module, EEPROM module, analogue comparator module, 16-bit and 8-bit timer/counter modules and pulse-width modulation (PWM). But before that, let's discuss the switches available on the ATmega32 development board and how to use these for specific applications.

Example 6: Input switches

On the ATmega32 development board, tactile (pushbutton) switches S1, S2, S3 and S4 are provided for input or debugging purpose. These switches are connected to Port-D pins PD0, PD1, PD2 and PD3, respectively. Of these, pins PD2 and PD3 are used by the external interrupt module to provide external interrupts. All the four switches are active-low type and hence don't affect the functioning of port pins otherwise.

To help you understand how to use these switches, a sample source code (switches.c) has been included in this month's EFY-CD. The source code lights up the LEDs connected to Port B in a to-and-fro sequential pattern. The sequential pattern or chaser effect can be stopped and restarted at any point of time using switches S2 and S1, respectively. Pressing of any of these switches is acknowledged by a beep sound produced by the buzzer on the board. The to-and-fro sequential patterns of the LEDs are produced using timer/counter0 module in the ATmega32. The timer is made to run in the

clear-timer-on-compare (CTC) mode as in the 8bit_timer.c code discussed in the first part of this article.

In the switches.c source code, first, you need to configure the port pins according to their purpose in the application. Here PD0 and PD1 are connected to

figured as input pins by clearing the corresponding bits in DDRD registers. The internal pull-up resistors for these pins are enabled by setting the corresponding bits in Port D register. Pin 21 (PD7) of ATmega32 is configured as an output pin with the buzzer connected to it.

De-bounce action of the input switches introduces an error. To eliminate this glitch, a de-bounce delay is introduced both when the pressing of a key is detected and when the key is read. By doing so, pressing of a valid key is registered. The de-bounce correction algorithm is implemented in the code using the following user-defined function:

```
unsigned int button(unsigned char
PORT, unsigned char PIN, unsigned int ...
TIME, unsigned char STATE);
```

Here input arguments PORT and PIN refer to the corresponding input/output (I/O) port and pin number used as the input pin, respectively. TIME refers to the de-bounce delay to be used and STATE refers to the active state of the switch connected to the input pin (here active state is 0).

Example 7: External interrupt module

Sample source code external_int.c will help you understand the working of the external interrupt module. This particular test code lights up the LEDs connected to Port B in sequential to-and-fro pattern. We have done this many times before, but here the rate of the LED chaser can be decreased and increased by a step of 50 ms each using switches S3 and S4, respectively. Also, the current rate of the LED chaser is displayed on the LCD module interfaced.

The external interrupts are trig-

PARTS LIST*Semiconductors:*

IC1	- ATmega32 microcontroller
IC2	- 24C04 EEPROM
IC3	- MAX232 RS-232 driver
IC4	- 7805, 5V regulator
ZD1, ZD2	- 5.1V zener diode
D1-D4	- 1N4007 rectifier diode
D5	- 1N4148 signal diode
LED1	- 5mm light emitting diode
LED2-LED9	- 3mm/flat LEDs
T1, T2	- BC548 npn transistor

Resistors (all ¼-watt, ±5% carbon)

R1-R9	- 470-ohm
R10	- 10-kilo-ohm
R11, R12, R16	- 8.2-kilo-ohm
R13	- 2.2-kilo-ohm
R14, R15	- 5-kilo-ohm
VR1, VR2	- 10-kilo-ohm preset

Capacitors:

C1	- 1000µF, 25V electrolytic
C2	- 0.1µF ceramic
C3	- 10µF, 16V electrolytic
C4, C5	- 22µF ceramic
C6-C10	- 1µF, 16V electrolytic

Miscellaneous:

S1-S5	- Tactile switch
S6	- On/off switch
S7, S8	- DPDT slide switch
X _{AL}	- 16MHz quartz crystal
CON1	- 2×8-pin berg strip male connector
CON2	- 3-pin male connector
Port A-Port D	- 8-pin berg strip male connector
PC Port	- 4-pin berg strip male connector
COM Port	- DB9 female PCB mountable connector
JS	- Shorting jumper
LCD	- 16×2 LCD module
	- DC connector
	- 40-pin DIL IC socket
	- 8-pin DIL IC socket
	- Piezobuzzer